

Full Length Article

Offline constrained policy optimization with safe anchoring

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ARTICLE INFO

Keywords:

Safe offline reinforcement learning
 Constrained policy optimization
 Behavior regularization
 Safe anchoring

ABSTRACT

The application of reinforcement learning (RL) in real-world scenarios is limited due to safety concerns and the distribution-shift challenge in the offline setting. To address these issues, we formulate safe offline RL as a constrained policy optimization problem that integrates cumulative cost constraints and behavioral policy regularization. We first derive the analytical solution of the offline constrained policy optimization problem through Lagrangian duality. Then, we prove that iterative updates of this solution guarantee monotonic performance improvement while bounding worst-case costs relative to the behavioral policy. To further prevent out-of-distribution actions that may violate safety constraints, we propose a mechanism that distills a “safe action” distribution from the offline data and restricts policy updates within this safe region. We term this approach safe anchoring. By projecting the analytical solution into a parameterized policy space using a VAE-distilled safe anchoring mechanism, we develop the Offline Constrained Policy Optimization with Safe Anchoring (OCPO-SA) algorithm. Extensive experiments on Safety-Gymnasium and Bullet-Safety-Gym demonstrate that OCPO-SA achieves safety in all tested environments, with the average cost reduced by 24% compared with the best-performing baseline among the compared algorithms.

1. Introduction

Reinforcement learning (RL) is a learning paradigm for sequential decision-making that enables agents to optimize long-term cumulative objectives through interaction with the environment. Moving beyond supervised or imitation learning, RL explicitly reasons about long-horizon rewards, demonstrating promising application prospects in various fields such as games (Silver et al., 2016), autonomous driving (Wang et al., 2023), robotics (Noh et al., 2025; Wang et al., 2024), and recommendation systems (Liu et al., 2025).

However, the deployment of RL in real-world scenarios is increasingly shifting towards safety-critical domains. In these contexts, the demand for secure and robust AI has highlighted the importance of reliable data-driven modeling. This is evident in applications such as IoT intrusion detection (Fares et al., 2025), medical diagnostics (Nafisah et al., 2025), and robust neural network state estimation (Cao et al., 2024; Subhashri et al., 2025). Consequently, ensuring safety in decision-making processes is no longer ancillary but a primary requirement. Nevertheless, a significant gap remains between the theoretical potential of RL and its practical deployment: the traditional trial-and-error exploration

mechanism often incurs prohibitive data collection costs and unacceptable risks of catastrophic failure.

To bridge this gap, RL must address two intertwined challenges: low sample efficiency and stringent safety concerns. To tackle both simultaneously, our work focuses on safe offline RL. This paradigm aims to learn constraint-satisfying policies from static datasets, thereby eliminating the need for dangerous online interactions while maximizing the utility of existing historical data. However, achieving this requires overcoming the inherent difficulties of both offline learning and safety-constrained optimization, as discussed below.

Offline reinforcement learning (offline RL) (Levine et al., 2020) improves sample efficiency by learning from a fixed dataset without additional environment interaction. However, offline RL is vulnerable to distribution-shift: the learned policy may induce state-action distributions that deviate from the dataset, leading to value overestimation and performance degradation (Huang et al., 2023). To mitigate this issue, many offline RL methods constrain policy updates to remain close to the behavior policy (e.g., via implicit behavior modeling (Fujimoto et al., 2019) or conservative value regularization (Kumar et al., 2020)).

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Safety is another critical requirement for deploying RL in real-world systems. Safe RL (Gu et al., 2022; Ray et al., 2019) is commonly modeled as a constrained Markov decision process (CMDP) (Altman, 2021), where constraints can be imposed as hard constraints at every step or as soft constraints in expectation over trajectories (Achiam et al., 2017; Chow et al., 2018; Tessler et al., 2018; Zhang et al., 2022).

Applying these safety mechanisms directly to static datasets, forming the safe offline RL paradigm, creates unique and intertwined complexities. While standard offline RL suffers from the distribution-shift problem (Levine et al., 2020), the strict safety requirements significantly amplify this mismatch vulnerability. Specifically, in standard offline RL, taking out-of-distribution (OOD) actions merely results in value overestimation and suboptimal rewards. In safe offline RL, however, the safety constraints are often more sensitive to distributional shifts; executing unknown OOD actions is highly likely to cause an underestimation of cost, leading to catastrophic safety violations in real-world deployment. Furthermore, because the agent is strictly prohibited from interacting with the environment to correct policy errors via trial-and-error, ensuring safety in the offline setting is inherently more challenging (Lee et al., 2022; Xu et al., 2022).

To address these challenges, existing methods like CPQ (Xu et al., 2022) and COptiDICE (Lee et al., 2022) model safe offline RL as a multi-constrained optimization problem (e.g., combining cost constraints and behavioral regularization). However, these works often lack a theoretical analytical solution for the offline constrained policy optimization problem, which limits the understanding of performance monotonicity. Moreover, in practice, these approaches may still fail to provide rigorous safety guarantees, with policies frequently violating cost constraints during deployment due to imperfect distributional control.

Motivated by the above discussion, we aim to develop a safe offline RL algorithm that bridges theoretical guarantees and practical safety. The main contributions of this work are listed as follows:

1. We derive an analytical solution to the safe offline constrained policy optimization problem using Lagrangian duality, proving that iterative policy updates ensure monotonic performance improvement and bound the worst-case cost relative to the behavioral policy.
2. We propose a safe anchoring mechanism using a conditional variational autoencoder (CVAE). We extract the distribution of safe actions from offline data to explicitly constrain policy updates, preventing out-of-distribution (OOD) actions that may violate safety constraints.
3. We introduce the OCPO-SA algorithm. Experimental results on the Safety-Gymnasium and Bullet-Safety-Gym benchmarks demonstrate that OCPO-SA achieves safety in all tested environments, with the average cost reduced by 24% compared with the best-performing baseline among the compared algorithms.

The remainder of this article is organized as follows. Section 2 reviews the related literature. Section 3 introduces the necessary preliminaries. Section 4 presents the proposed algorithm and theoretical analysis. Section 5 evaluates the method through experiments, and Section 6 provides concluding remarks.

2. Related work

2.1. Offline RL

Offline RL aims to learn RL policies from pre-collected datasets. The primary challenge faced by offline RL is the distribution-shift (Fujimoto et al., 2019), wherein the distribution of learned policies does not match that of the behavioral policy. Batch constrained Q-learning (BCQ) (Fujimoto et al., 2019) is an offline algorithm that can be used for tasks with continuous state-action space. It also does not require knowledge of the specific behavioral policy that generates the training data, but instead learns a representation of that behavioral policy by variational autoencoders (VAE), which can in turn be used to sample from the behavioral

policy to implement policy constraints on the current policy. Bootstrapping error cumulative reduction (BEAR) (Kumar et al., 2019) introduces the actor-critic idea in offline RL settings with a narrow data distribution. It learns the policy by penalizing the Maximum Mean Discrepancy (MMD) distance between the actions performed by the updated policy and the actions sampled from the behavioral policy. Further, Fujimoto and Gu (2021) introduced TD3-BC, a simple and efficient algorithm that does not include behavioral policy modeling. Our work refers to the idea of policy constraint method to alleviate the distribution-shift problem in the safe offline RL setting.

Recently, generative-model-based offline RL methods have been proposed to better capture complex actions (Wang et al., 2022) or trajectory distributions (Janner et al., 2022) from offline datasets. These approaches typically treat offline RL as a conditional generative modeling problem and can improve long-horizon planning by sampling and refining candidate trajectories. While such diffusion-based methods are promising for offline RL, they are not directly designed for CMDPs with explicit cost constraints, which motivates the need for safe offline RL methods with principled constraint handling.

2.2. Safe offline RL

In the safe RL setting, the objective is to learn a policy to maximize the long-term return and ensure the safety constraint. For this purpose, the Lagrangian-based model-free methods in Chen et al. (2021b) and Tessler et al. (2018) added a penalty term for constraint violation to the expected return target and solved the resulting optimization problem through learnable Lagrangian multipliers. However, the Lagrangian-based method can only asymptotically satisfy the constraints and cannot guarantee safety during the training process when interacting with the real environment. On the other hand, this method is sensitive to the initialization and learning rate of the Lagrangian multiplier. Stooke et al. (2020) proposed to use the PID regulation law to update the Lagrangian coefficient, which effectively alleviated the oscillation problem during training. The constrained policy optimization (CPO) method (Achiam et al., 2017) extends the trust region optimization idea (Schulman et al., 2015), which can satisfy the constraints during training, but the computational cost will increase significantly if there are multiple constraints. Note that these online safe RL algorithms cannot be applied to the offline setting.

In the safe offline RL setting, it is natural to introduce cost constraints into traditional offline algorithms and use primal-dual optimization to solve the resulting constrained problems (Chen et al., 2021b), such as BCQ-lag (Fujimoto et al., 2019) and BEAR-lag (Kumar et al., 2019). CPQ (Xu et al., 2022) addresses safety by pessimistically estimating the cost value function of out-of-distribution actions and penalizing constraint violations during policy training. Diverging from such approaches, COptiDICE (Lee et al., 2022) bypasses explicit policy or value function modeling by directly optimizing the state-action stationary distribution and then recovering the policy through importance-weighted behavioral cloning. Similarly, CDT (Liu et al., 2023b) combines decision transformers (Chen et al., 2021a) with data augmentation and formulates the problem from a multi-objective optimization perspective. Building on these foundations, more recent studies have advanced safe offline RL along different directions: Lin et al. (2023) proposed a framework with real-time budget constraints to enforce resource limitations, Zheng et al. (2024) introduced a feasibility-guided diffusion model leveraging generative modeling to improve both safety and performance, Yao et al. (2024) developed OASIS, which adaptively shapes conditional distributions to constrain policy updates, and Yu et al. (2022) explored reachability-constrained reinforcement learning by incorporating reachability analysis into constrained optimization.

Despite these advancements, existing methods often rely on numerical approximations, lacking closed-form analytical solutions or rigorous performance monotonicity analysis. Consequently, they frequently fail to ensure strict safety, leading to cost violations in deployment. To

bridge this gap, we formulate safe offline RL using both cost and behavioral constraints. By theoretically analyzing the behavioral policy's impact, we derive an analytical solution that guarantees monotonic improvement and robust safety via a novel safe anchoring mechanism.

3. Preliminary

The safe RL problem is modeled as a CMDP which is defined by a six-tuple $\{S, A, P, r, c, \gamma\}$, where S represents a finite set of states; A represents a finite set of actions; $P : S \times A \times S \rightarrow [0, 1]$ is the state transition probability function; $r : S \times A \rightarrow \mathbb{R}$ is the reward function; $c : S \times A \rightarrow \mathbb{R}$ is the cost function; $\gamma \in [0, 1)$ is the discount factor. The objective of safe RL is to find a policy that maximizes the expected cumulative reward while satisfying a cost constraint condition

$$\begin{aligned} \max_{\pi} J(\pi) &= \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \\ \text{s.t. } J_c(\pi) &= \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t) \right] \leq d, \end{aligned} \quad (1)$$

where $\tau = \{(s_t, a_t, r_t, c_t)\}_{t=0}^T$ is the trajectory obtained by the interaction between the agent and the environment, $\tau \sim \pi$ means that trajectory τ is generated by executing policy π , and d is the restricted threshold of the expected cumulative cost. For the policy π , the action-value function, state-value function, and advantage function are defined as:

$$Q^{\pi}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^{\pi}(s')],$$

$$V^{\pi}(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} [Q^{\pi}(s, a)],$$

and

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s).$$

Similarly, the above concepts for the cost function $c(s, a)$ are defined as:

$$Q_c^{\pi}(s, a) = c(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_c^{\pi}(s')],$$

$$V_c^{\pi}(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} [Q_c^{\pi}(s, a)],$$

and

$$A_c^{\pi}(s, a) = Q_c^{\pi}(s, a) - V_c^{\pi}(s).$$

In the safe offline RL setting, it is assumed that there is no interaction between the agent and the environment. The agent only learns policy from an offline dataset $\mathcal{D} = \{s, a, s', r(s, a), c(s, a)\}$, generated by unknown arbitrary behavior policies. Since the behavioral policy is unknown, we use imitation learning method to imitate the behavior action of each state by supervised learning:

$$\hat{\pi}_{\beta} = \underset{\pi}{\operatorname{argmax}} \mathbb{E}_{(s, a) \sim \mathcal{D}} [\log \pi(a|s)].$$

The goal of safe offline RL is to learn a policy by using $\mathcal{D}$ that maximizes the cumulative reward while satisfying the cumulative cost constraint in (1). Note that the offline dataset may contain the trajectories that violate safety constraints. In addition, the distribution-shift is the main challenge for offline learning, including safe offline learning.

4. Offline constrained policy optimization

4.1. Analysis of offline conservative policy improvement

In order to prevent distribution-shift in the safe offline RL setting, a natural idea is to introduce the behavior regularization mechanism into the constrained policy optimization. Similar to Lee et al. (2022), Xu et al. (2022), the following offline conservative policy improvement is considered:

$$\pi_{k+1} = \underset{\pi}{\operatorname{argmax}} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi(\cdot|s)} [Q^{\pi_k}(s, a)] \quad (2a)$$

$$\text{s.t. } \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi(\cdot|s)} [Q_c^{\pi_k}(s, a)] \leq d, \quad (2b)$$

$$\mathbb{E}_{s \sim \mathcal{D}} [D_{\text{KL}}(\pi(\cdot|s) \|\hat{\pi}_{\beta}(\cdot|s))] \leq \rho, \quad (2c)$$

$$\int_a \pi(a|s) da = 1, \quad (2d)$$

where π_k represents the iterative policy and k denotes the iteration index; $D_{\text{KL}}(\pi(\cdot|s) \|\hat{\pi}_{\beta}(\cdot|s)) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\log \frac{\pi(a|s)}{\hat{\pi}_{\beta}(a|s)} \right]$ is the KL divergence between $\pi(\cdot|s)$ and $\hat{\pi}_{\beta}(\cdot|s)$; ρ is a positive constant that characterizes the tightness of policy constraint. The constraint (2b) bounds the expected cost. Similar to BRAC (Wu et al., 2019), the constraint (2c) uses KL divergence to keep the updated policy π_{k+1} close to the behavioral policy $\hat{\pi}_{\beta}$, ensuring stable policy updates and mitigating distribution-shift. Here, KL divergence is chosen for its theoretical tractability (enabling closed-form solutions). The constraint (2d) ensures $\pi(\cdot|s)$ is a valid probability distribution.

Theorem 1. The analytical solution for (2) is

$$\pi_{k+1}^*(a|s) = \frac{\hat{\pi}_{\beta}(a|s)}{Z(s)} \exp \left(\frac{Q^{\pi_k}(s, a) - \lambda^* Q_c^{\pi_k}(s, a)}{v^*} \right), \quad (3)$$

where $Z(s) = \int_a \hat{\pi}_{\beta}(a|s) \exp \left(\frac{Q^{\pi_k}(s, a) - \lambda^* Q_c^{\pi_k}(s, a)}{v^*} \right) da$ is the normalizer ensuring $\pi_{k+1}^*(\cdot|s)$ is a valid distribution, and the dual variables λ^*, v^* solve the convex problem

$$\min_{\lambda, v \geq 0} g(v, \lambda) = \lambda d + v \rho + v \mathbb{E}_{\pi_{\beta}} \left[\log \mathbb{E}_{\pi_k} \left(\exp \left(\frac{Q^{\pi_k}(s, a) - \lambda Q_c^{\pi_k}(s, a)}{v} \right) \right) \right]. \quad (4)$$

Proof. The analytic solution for the constrained optimization problem (2) can be obtained by enforcing the KKT condition. We first convert (2) to its equivalent Lagrangian function:

$$\begin{aligned} \mathcal{L}(\pi, \lambda, v, \alpha) &= \mathbb{E}_{a \sim \pi(\cdot|s)} [Q^{\pi_k}(s, a) + \lambda(d - Q_c^{\pi_k}(s, a)) \\ &\quad + v(\rho - D_{\text{KL}}(\pi(\cdot|s) \|\hat{\pi}_{\beta}(\cdot|s))) + \alpha(1 - \int_a \pi(a|s) da)], \end{aligned} \quad (5)$$

where λ, v , and α are the Lagrange multipliers for the constraints. Given that the objective function is linear and all imposed constraints adhere to convexity, the constrained optimization problem (2) is convex. Then we obtain the following equivalent dual problem:

$$\min_{\lambda, v, \alpha} \mathcal{L}(\pi, \lambda, v, \alpha). \quad (6)$$

Differentiating $\mathcal{L}(\pi, \lambda, v, \alpha)$ with respect to $\pi(a|s)$ gives:

$$\frac{\partial \mathcal{L}}{\partial \pi(a|s)} = Q^{\pi_k}(s, a) - \lambda Q_c^{\pi_k}(s, a) + v \log \hat{\pi}_{\beta}(a|s) - v \log \pi(a|s) - v - \alpha. \quad (7)$$

Setting $\frac{\partial \mathcal{L}}{\partial \pi(a|s)}$ to zero gives the closed form solution:

$$\begin{aligned} \pi_{k+1}^*(a|s) &= \frac{\hat{\pi}_{\beta}(a|s)}{\exp(\frac{v+\alpha}{v})} \exp \left(\frac{Q^{\pi_k}(s, a) - \lambda Q_c^{\pi_k}(s, a)}{v} \right) \\ &= \frac{\hat{\pi}_{\beta}(a|s)}{Z(s)} \exp \left(\frac{Q^{\pi_k}(s, a) - \lambda Q_c^{\pi_k}(s, a)}{v} \right), \end{aligned} \quad (8)$$

where $Z(s) = \exp(\frac{v+\alpha}{v})$ could be viewed as a normalizer for $\pi(\cdot|s)$ since it is a constant that is independent of $\pi(a|s)$. To solve for λ, v , and α , taking the closed form solution π_{k+1}^* back to the Lagrangian function (5), we get

$$\begin{aligned} \mathcal{L}(\pi_{k+1}^*, \lambda, v, \alpha) &= \mathbb{E}_{a \sim \pi_{k+1}^*(\cdot|s)} [Q^{\pi_k}(s, a) + \lambda(d - Q_c^{\pi_k}(s, a)) \\ &\quad + v(\rho - D_{\text{KL}}(\pi_{k+1}^*(a|s) \|\hat{\pi}_{\beta}(a|s))) + \alpha(1 - \int_a \pi_{k+1}^*(a|s) da)] \\ &= \lambda d + v(\rho - D_{\text{KL}}(\pi_{k+1}^*(a|s) \|\hat{\pi}_{\beta}(a|s))). \end{aligned}$$

One can find that most of the terms are cancelled out, leading to the dual function:

$$g(v, \lambda) = \lambda d + v \rho + v \mathbb{E}_{\hat{\pi}_{\beta}} \left[\log \mathbb{E}_{\pi_k} \left[\exp \left(\frac{Q^{\pi_k}(s, a) - \lambda Q_c^{\pi_k}(s, a)}{v} \right) \right] \right]. \quad (9)$$

In (9), the dual function $g(v, \lambda)$ does not explicitly depend on α because the optimal policy π_{k+1}^* in (3) inherently satisfies the normalization constraint $\int \pi_{k+1}^*(a|s)da = 1$. The optimal dual variables are calculated by

$$\lambda^*, v^* = \arg \min_{\lambda, v \geq 0} g(\lambda, v). \quad (10)$$

□

In Theorem 1, the constrained optimization problem (2) is transformed into a dual optimization problem to obtain the analytic solution. It is known from Theorem 1 that the problem (2) can be solved in an optimal form with non-parametric solution, as we ensure zero duality gap. Eq. (3) illustrates that the optimal policy is a weighted combination of the behavioral policy, with weights controlled by Q^{π_k} and $Q_c^{\pi_k}$. It is observed that higher weights are assigned to higher returns and lower costs, with a balanced trade-off parameter λ . Intuitively, the dual variable v serves as a temperature regulator, moderating the uniformity of weights. According to Theorem 1, the constraint parameters d and ρ are implicitly integrated through solving for the optimal Lagrange multipliers λ and v , thus eliminating the need for their explicit appearance in the final optimal policy expression. Existing studies such as Lee et al. (2022), Xu et al. (2022) introduce optimization problems similar to (2) but do not provide analytical solutions. In contrast, Theorem 1 provides a theoretical analytical solution via Lagrangian duality.

Remark 1. To provide an intuitive understanding, the optimization in Eq. (2) seeks a policy that optimally balances reward maximization, strict safety satisfaction, and behavioral regularization based solely on static data. Unlike purely numerical gradient-based approximations, our analytical solution in Eq. (3) yields an exact, closed-form mathematical structure for this optimal policy. For example, consider a robot navigating an intersection where the historical dataset favors a fast but risky path over a slow but safe one. Rather than blindly mimicking the risky historical behavior, the analytical solution explicitly re-calibrates the action distribution by exponentially penalizing the risky action based on its safety cost and boosting the safe action. This ensures the theoretically optimal policy strictly adheres to safety bounds before being projected into the neural network space.

Theorem 2. Let ϵ_{shift} be the maximum expected error induced by the distributional mismatch between the empirical dataset $\mathcal{D}$ and the true state visitation distribution. The performance of the policy sequence obtained by (2) satisfies an approximate monotonic improvement, i.e., $J(\pi_{k+1}) \geq J(\pi_k) - \epsilon_{shift}$, for $k = 0, 1, 2, \dots$

Proof. Let $\Theta = \{\pi(\cdot|s) | Q_c^\pi(s, a) \leq d, D_{KL}[\pi(\cdot|s) \| \hat{\pi}_\beta(\cdot|s)] \leq \rho, \forall s \in S, \forall a \in A\}$. According to (2), we have

$$\pi_{k+1}(\cdot | s) = \arg \max_{\pi(\cdot|s) \in \Theta} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi(\cdot|s)} [Q^{\pi_k}(s, a)]. \quad (11)$$

Because the optimization is performed over the empirical dataset $\mathcal{D}$ rather than the true state distribution, the point-wise improvement holds up to a local distributional shift error $\epsilon(s)$, i.e.,

$$\mathbb{E}_{a \sim \pi_{k+1}(\cdot|s)} [Q^{\pi_k}(s, a)] \geq \mathbb{E}_{a \sim \pi_k(\cdot|s)} [Q^{\pi_k}(s, a)] - \epsilon(s), \quad \forall s \in S. \quad (12)$$

Thus, for any state-action pair (s_t, a_t) , expanding the Q -value yields

$$\begin{aligned} Q^{\pi_k}(s_t, a_t) &= r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot|s_t, a_t)} [V^{\pi_k}(s_{t+1})] \\ &\leq r(s_t, a_t) + \gamma \mathbb{E}_{\substack{s_{t+1} \sim P(\cdot|s_t, a_t) \\ a_{t+1} \sim \pi_{k+1}(\cdot|s_{t+1})}} [Q^{\pi_k}(s_{t+1}, a_{t+1})] + \gamma \mathbb{E}[\epsilon(s_{t+1})] \\ &\leq r(s_t, a_t) + \gamma \mathbb{E}_{\substack{s_{t+1} \sim P(\cdot|s_t, a_t) \\ a_{t+1} \sim \pi_{k+1}(\cdot|s_{t+1})}} [r(s_{t+1}, a_{t+1})] \\ &\quad + \gamma \mathbb{E}_{s_{t+2} \sim P(\cdot|s_{t+1}, a_{t+1})} [V^{\pi_k}(s_{t+2})] \\ &\quad + \gamma \mathbb{E}[\epsilon(s_{t+1})] + \gamma^2 \mathbb{E}[\epsilon(s_{t+2})] \\ &\vdots \\ &\leq r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot|s_t, a_t)} [V^{\pi_{k+1}}(s_{t+1})] + \sum_{i=1}^{\infty} \gamma^i \mathbb{E}[\epsilon(s_{t+i})] \end{aligned}$$

$$= Q^{\pi_{k+1}}(s_t, a_t) + \epsilon_{shift}(s_t, a_t). \quad (13)$$

Taking the expectation over the initial state distribution, we obtain $J(\pi_{k+1}) \geq J(\pi_k) - \epsilon_{shift}$, where ϵ_{shift} is the accumulated mismatch penalty. □

Theorem 2 indicates that employing (2) to update the policy ensures that the performance of the learned policy π_{k+1} surpasses that of the old policy π_k , thus ensuring the monotonicity of policy performance. This provision assures the convergence of policies towards the optimal policy.

Theorem 3. Consider (2), the worst-case cost bound of π_{k+1} is

$$J_c(\pi_{k+1}) \leq J_c(\hat{\pi}_\beta) + \frac{\mathcal{A}_c^{\hat{\pi}_\beta}(1 - \gamma + \gamma\sqrt{2\rho})}{(1 - \gamma)^2}, \quad (14)$$

where $\mathcal{A}_c^{\hat{\pi}_\beta} = \max_{s,a} |A_c^{\hat{\pi}_\beta}(s, a)|$.

Proof. According to Achiam et al. (2017), the following relationship holds for any two policy π' and π ,

$$\begin{aligned} J_c(\pi') - J_c(\pi) &\leq \frac{1}{1 - \gamma} \mathbb{E}_{\substack{s \sim \rho_\pi(\cdot) \\ a \sim \pi'(\cdot|s)}} [A_c^\pi(s, a)] \\ &\quad + \frac{2\gamma\mathcal{A}_c^{\pi'}}{(1 - \gamma)^2} \sqrt{\frac{1}{2} \mathbb{E}_{s \sim \rho_\pi(\cdot)} [D_{KL}[\pi'(\cdot|s) \| \pi(\cdot|s)]]}, \end{aligned} \quad (15)$$

where $\rho^\pi(s)$ is the stationary distributions of states under policy $\pi(\cdot|s)$ and $\mathcal{A}_c^{\pi'} = \max_{s,a} |A_c^{\pi'}(s, a)|$. This shows that

$$\begin{aligned} J_c(\pi_{k+1}) - J_c(\hat{\pi}_\beta) &\leq \frac{1}{1 - \gamma} \mathbb{E}_{\substack{s \sim \rho_{\hat{\pi}_\beta}(\cdot) \\ a \sim \pi_{k+1}(\cdot|s)}} [A_c^{\hat{\pi}_\beta}(s, a)] \\ &\quad + \frac{2\gamma\mathcal{A}_c^{\hat{\pi}_\beta}}{(1 - \gamma)^2} \sqrt{\frac{1}{2} \mathbb{E}_{s \sim \rho_{\hat{\pi}_\beta}(\cdot|s)} [D_{KL}[\pi_{k+1}(\cdot|s) \| \hat{\pi}_\beta(\cdot|s)]]} \\ &\leq \frac{\mathcal{A}_c^{\hat{\pi}_\beta}(1 - \gamma + \gamma\sqrt{2\rho})}{(1 - \gamma)^2}. \end{aligned} \quad (16)$$

From (16), we have

$$J_c(\pi_{k+1}) \leq J_c(\hat{\pi}_\beta) + \frac{\mathcal{A}_c^{\hat{\pi}_\beta}(1 - \gamma + \gamma\sqrt{2\rho})}{(1 - \gamma)^2}. \quad (17)$$

□

Theorem 3 elucidates that employing (2) to update the policy ensures that the expected cost of the obtained policy remains bounded, thereby providing assurance for the safety of the policy. Eq. (2) establishes a fundamental relationship between the cost incurred by the learned policy and that of the behavioral policy. This equation illustrates how the worst cost bound associated with the learned policy is intrinsically linked to the behavioral policy, highlighting a pivotal feature of safe offline RL. In fact, it is necessary to consider the nature of the behavior observed during the training data collection phase. Drawing on Theorems 2 and 3, we conclude that by iterating (2), a sequence of policies characterized by monotonic performance improvement and bounded expected costs can be obtained. This aligns precisely with the objectives sought in safe offline RL.

4.2. Offline constrained policy optimization with safe anchoring

In Section 4.1, we analyzed the monotonicity of policy performance and upper bounds of the cost of the learned policy under the offline conservative policy improvement. Based on the theoretical results, we shall develop a practical safe offline RL algorithm named offline constrained policy optimization with safe anchoring.

In OCPO-SA, both the action-value and action-cost functions and the policy are approximated by neural networks with parameters ϕ , ϕ_c and

θ respectively. The loss functions for the Q-function network and the Qc-function network are

$$L(\phi) = \mathbb{E}_{(s,a,r,s') \sim D} \left[(y_r - Q_\phi(s, a))^2 \right] \quad (18)$$

and

$$L(\phi_c) = \mathbb{E}_{(s,a,r,s') \sim D} \left[(y_c - Q_{\phi_c}(s, a))^2 \right], \quad (19)$$

where the target values are

$$y_r = r(s, a) + \gamma Q_{\bar{\phi}}(s', a' \sim \pi_{\bar{\theta}}(\cdot|s')),$$

$$y_c = c(s, a) + \gamma Q_{\bar{\phi}_c}(s', a' \sim \pi_{\bar{\theta}}(\cdot|s')),$$

$Q_{\bar{\phi}}$ and $Q_{\bar{\phi}_c}$ represent the target Q-function and target Qc-function networks, $\pi_{\bar{\theta}}$ represents the target policy network.

We have provided the analytical solution for the offline conservative policy improvement (2) in Theorem 1. Since the constant normalizer $Z(s)$ in (3) is unknown, we project this optimal non-parametric solution into the policy space by minimizing the KL divergence between the policy $\pi(\cdot|s)$ and $\pi_{k+1}^*(\cdot|s)$ under the state contained in D , i.e.,

$$\pi_{k+1}(\cdot|s) = \arg \min_{\pi} \mathbb{E}_{s \sim D} [D_{KL}(\pi(\cdot|s) \| \pi_{k+1}^*(\cdot|s))]. \quad (20)$$

In OCPO-SA, the policy parameter θ is updated by minimizing the KL divergence between π_{θ} and the optimal non-parametric policy π_{k+1}^* . Mathematically, the policy loss is expressed as:

$$\begin{aligned} L(\theta) &= \mathbb{E}_{s \sim D} [D_{KL}(\pi_{\theta}(\cdot|s) \| \pi_{k+1}^*(\cdot|s))] \\ &= \mathbb{E}_{s \sim D, a \sim \pi_{\theta}(\cdot|s)} \left[\log \frac{\pi_{\theta}(a|s)}{\pi_{k+1}^*(a|s)} \right]. \end{aligned} \quad (21)$$

Substituting the optimal solution (3) into (21), we derive the policy loss function:

$$L(\theta) = \mathbb{E}_{s \sim D, a \sim \pi_{\theta}(\cdot|s)} \left[-\frac{Q_{\phi}(s, a)}{v^*} + \frac{\lambda^* Q_{\phi_c}(s, a)}{v^*} + D_{KL}(\pi_{\theta}(\cdot|s) \| \pi_{\beta}(\cdot|s)) \right]. \quad (22)$$

For practical implementation, we consider deterministic policies. If $\pi_{\theta}(\cdot|s)$ and $\pi_{\beta}(\cdot|s)$ follow multivariate Gaussian distributions with identity covariance matrices, we can obtain

$$D_{KL}(\pi_{\theta}(\cdot|s) \| \pi_{\beta}(\cdot|s)) = \frac{1}{2} \|\hat{\mu}_1 - \hat{\mu}_2\|^2, \quad (23)$$

where $\hat{\mu}_1$ and $\hat{\mu}_2$ are the mean vectors of $\pi_{\theta}(\cdot|s)$ and $\pi_{\beta}(\cdot|s)$. Substituting (23) into (22), the policy loss (22) becomes

$$L(\theta) = \mathbb{E}_{(s,a) \sim D} \left[-\frac{Q_{\phi}(s, \pi_{\theta}(s))}{v^*} + \frac{\lambda^* Q_{\phi_c}(s, \pi_{\theta}(s))}{v^*} + \|\pi_{\theta}(s) - a\|^2 \right]. \quad (24)$$

To address potential instability from varying Q_{ϕ} and Q_{ϕ_c} magnitudes, we adopt the normalization scheme inspired by TD3_BC (Fujimoto & Gu, 2021). The normalized policy objective function is

$$L(\theta) = \mathbb{E}_{(s,a) \sim D} \left[-\eta_1 Q_{\phi}(s, \pi_{\theta}(s)) + \eta_2 \lambda^* Q_{\phi_c}(s, \pi_{\theta}(s)) + \|\pi_{\theta}(s) - a\|^2 \right], \quad (25)$$

where the coefficients η_1 and η_2 are defined as:

$$\eta_1 = \frac{\alpha_1}{\frac{1}{N} \sum_{(s,a)} |Q_{\phi}(s, a)|}, \eta_2 = \frac{\alpha_2}{\frac{1}{N} \sum_{(s,a)} |Q_{\phi_c}(s, a)|}. \quad (26)$$

Here, α_1 and α_2 serve as global scaling factors to balance the return objective and cost constraint, N is the mini-batch size. This normalization stabilizes Q-value scales. Note that the Lagrangian multipliers v^* is incorporated into α_1 and α_2 .

Theorem 3 indicates that the safety of the policy learned by OCPO-SA is influenced by the behavioral policy π_{β} . If there are unsafe trajectories in D , it can be difficult to learn a policy that satisfies the cost constraint. To eliminate potentially unsafe actions contained in offline dataset D , we design a safe anchoring mechanism. The proposed safe anchoring comprises three components: Safe data filtering, safe policy distillation, and dual safety-constrained policy optimization.

4.2.1. Safe data filtering

Based on the trajectories in the offline dataset D , we construct a safe dataset as follows:

$$D_{\text{safe}} = \left\{ \tau \mid \sum_{t=0}^T \gamma^t c_t \leq d_{\text{safe}} \wedge \tau \in D \right\}, \quad (27)$$

where $\tau = \{(s_t, a_t, r_t, c_t)\}_{t=0}^T$ and d_{safe} is the predefined safety threshold. In OCPO-SA, d_{safe} is set as d .

4.2.2. Safe policy distillation

We train a safe behavioral policy $\mu_{\beta}^{\text{safe}}$ on D_{safe} using a CVAE. Following Fujimoto et al. (2019), the objective function for training the CVAE is:

$$\mathcal{L}_{\text{CVAE}} = \mathbb{E}_{(s,a) \sim D_{\text{safe}}} \left[\|a - \mu_{\phi_{\beta}}^{\text{safe}}(s)\|^2 + D_{KL}(q_{\phi_e}(z|s, a) \| p(z)) \right]. \quad (28)$$

Here, q_{ϕ_e} is the VAE encoder, $\mu_{\phi_{\beta}}^{\text{safe}}(s)$ is the mean of safe actions generated by the decoder parameterized by ϕ_{β} , z is the latent variable.

4.2.3. Dual safety-constrained policy optimization

Algorithm 1 OCPO-SA.

Input: $D = \{(s, a, r, c, s')\}_N$, weighting factors α_1, α_2 , target network smoothing parameter ψ , cost threshold d , training steps T_{max} , safety threshold d_{safe} , CVAE training epochs E_{CVAE}

Initialize: Actor network parameter θ , critic network parameters ϕ , cost critic network parameters ϕ_c , target network parameter $\bar{\theta} \leftarrow \theta$, CVAE parameters ϕ_{β} , $\bar{\phi} \leftarrow \phi$ and $\bar{\phi}_c \leftarrow \phi_c$, Lagrange multiplier λ , PID controller parameters K_p, K_D , and K_I

Safe Data Filtering:

$D_{\text{safe}} \leftarrow \emptyset$

for each trajectory $\tau = \{(s_t, a_t, c_t)\}_{t=0}^T$ in D **do**

$\text{cost} \leftarrow \sum_{t=0}^T \gamma^t c(s_t, a_t)$

if $\text{cost} \leq d_{\text{safe}}$ **then**

$D_{\text{safe}} \leftarrow D_{\text{safe}} \cup \{\tau\}$

end if

end for

Safe Policy Distillation:

for $h = 0$ to E_{CVAE} **do**

Sample a mini-batch $\bar{D}_{\text{safe}}$ from D_{safe}

Update ϕ_{β} via minimizing the CVAE loss (28)

end for

for $l = 0$ to T_{max} **do**

Sampling a mini-batch $\bar{D}$ from D

Update ϕ via minimizing the critic loss (18)

Update ϕ_c via minimizing the cost critic loss (19)

Calculate $e_l = Q_{\phi_c} - d$

Update PID Lagrange multiplier λ by (30).

Update θ via maximizing the policy loss (29)

Update target network parameters by

$$\bar{\theta} \leftarrow (1 - \psi)\bar{\theta} + \psi\theta$$

$$\bar{\phi} \leftarrow (1 - \psi)\bar{\phi} + \psi\phi$$

$$\bar{\phi}_c \leftarrow (1 - \psi)\bar{\phi}_c + \psi\phi_c$$

end for

Based on the safe behavioral policy $\mu_{\beta}^{\text{safe}}$, we modify the policy loss function as:

$$L(\theta) = \mathbb{E}_{(s,a) \sim D} \left[-\eta_1 Q_{\phi}(s, \pi_{\theta}(s)) + \lambda^* \eta_2 Q_{\phi_c}(s, \pi_{\theta}(s)) + \|\pi_{\theta}(s) - \mu_{\phi_{\beta}}^{\text{safe}}(s)\|^2 \right]. \quad (29)$$

This design incorporates dual safety mechanisms: 1) Constraining policy updates within the safe action space centered around $\mu_{\phi_{\beta}}^{\text{safe}}(s)$ via $\|\pi_{\theta}(s) - \mu_{\phi_{\beta}}^{\text{safe}}(s)\|^2$; 2) Penalizing high-risk actions through the $\lambda^* Q_{\phi_c}$ term, where λ^* is determined by the analytical solution in Theorem 1.

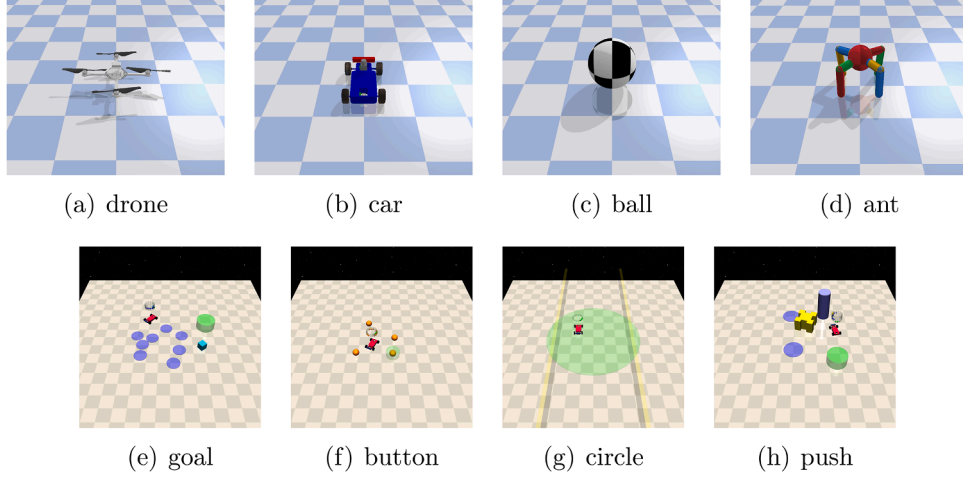


Fig. 1. Screenshots of the test platform tasks. (a) Drone, (b) Car, (c) Ball, (d) Ant, (e) Goal, (f) Button, (g) Circle, and (h) Push.

As for the Lagrangian multiplier λ^* is used to balance the trade-off between the reward maximization and the satisfaction of the cost constraint. Considering that traditional Lagrangian methods for constrained optimization often suffer from oscillatory dynamics and overshoot, leading to constraint violations in safe RL, the PID-Lagrangian framework was proposed in Stooke et al. (2020). We adapt this framework to offline settings and posit that the PID regulation can enhance training stability and accelerate convergence by dynamically adjusting Lagrangian multiplier based on constraint function. At the training step l , the update rule for λ_l is given by:

$$\lambda_l = K_p e_l + K_D (e_l - e_{l-1}) + K_I \sum_{i=0}^l e_i, \quad (30)$$

where $e_l = (\sum_{i=0}^{l-1} \gamma^i c_i + \gamma^l Q_{\phi_c}(s_l, a_l)) - d$. This error formulation aligns the signal with the CMDP definition by combining the actual accumulated costs to go $\sum_{i=0}^{l-1} \gamma^i c_i$ up to time step $l-1$ and the expected future cost-to-go Q_{ϕ_c} against the episodic threshold d . The three coefficients K_p , K_i , and K_d play different roles in the PID control mechanism. K_p is the proportional coefficient. It determines the immediate response of the system to the current constraint violation. A larger K_p will make the system respond more quickly to the current error, but it may also cause over-correction. K_i is the integral coefficient. It accumulates the past constraint violations over time. K_d is the derivative coefficient. It is based on the rate of change of the constraint violation. K_d can predict the future trend of the error and help to dampen oscillations. The implementation details of OCPO-SA are presented in the pseudo-code of Algorithm 1.

5. Experiments

In this section, we first outline our experimental settings, including baselines, benchmarks, metrics, and implementation details. Then, we present our experimental results and analysis. In experiments, we verify that the proposed method effectively outperforms other popular comparative methods on various popular safe RL tasks. The configurations of our implementation are listed as follows. Regarding software configurations, we used the CUDA 11.8, Torch 1.13.1, bullet-safety-gym 1.4.0, and safety-gymnasium 0.4.0. For hardware configuration, we employed an AMD Ryzen Threadripper PRO 5995WX 64-Cores processor, 128 GB RAM, and two NVIDIA RTX 3090Ti GPUs, running on a device with Ubuntu 20.04.4 LTS.

5.1. Implementation details

We conducted comprehensive evaluations on two widely used safe offline RL benchmarks integrated into the DSRL (Liu et al., 2023a) platform: Safety-Gymnasium (Ji et al., 2023; Ray et al., 2019) and Bullet-Safety-Gym (Gronauer, 2022). DSRL provides standardized datasets, evaluation protocols, and open-source implementations for 38 safe RL tasks across diverse domains, including robot control and autonomous driving. All reported benchmark results in this work are derived from DSRL's official datasets and evaluation pipelines, ensuring reproducibility and fairness. For scenarios not explicitly listed in DSRL's original tables (e.g., BallRun and CarRun), we followed its open-source codebase to generate results. A screenshot of the tasks are shown in Fig. 1. The details of the test platforms are described as follows

1. Bullet-Safety-Gym has four agents (Ball, Car, Drone, Ant) for tasks (Circle, Gather, Reach, Run) involving navigation/collection/goal-reaching with hazard avoidance, using rewards/costs for safe performance.
2. Safety-Gymnasium tasks guide Point/Car robots to green goals (red robots, purple hazards); our experiments use the Car agent.

The offline dataset D is obtained from the DSRL benchmark (Liu et al., 2023a), which is constructed to reflect the complexities of real-world data, characterized by diverse safety distributions and noise levels. The trajectories in DSRL are collected using a mixture of behavior policies with varying expertise levels: *Random*, *Medium*, and *Expert*. This mixture introduces noise through stochastic exploration and sub-optimal decision-making. As visualized in the reward-cost grid in Fig. 2, the datasets exhibit a wide spectrum of cost returns, covering both safe (low-cost) regions and high-risk (high-cost) zones. This diversity ensures that the offline agent must explicitly reason about the trade-off between maximizing rewards and satisfying safety constraints within a heterogeneous data distribution.

We introduce experimental evaluation metric calculations. For task $\mathcal{M}$, $r_{\max}(\mathcal{M})$ and $r_{\min}(\mathcal{M})$ are maximum and minimum returns. The normalized reward is $R_{\text{normalized}} = \frac{R_{\pi} - r_{\min}(\mathcal{M})}{r_{\max}(\mathcal{M}) - r_{\min}(\mathcal{M})}$, where R_{π} is the return of policy π . In safe RL, fixed bounds are used due to post-processing impacts on dataset quality.

The normalized cost is defined as $C_{\text{normalized}} = \frac{C_{\pi} + \epsilon}{d + \epsilon}$, where C_{π} is the cost return, d is a predefined threshold, and ϵ is a small positive constant for numerical stability when $d = 0$.

The hyper-parameters of OCPO-SA are presented in Table 1. To demonstrate the superior performance of OCPO-SA, the OCPO-SA is benchmarked against a diverse array of state-of-the-art baseline algorithms. The comparative algorithms include:

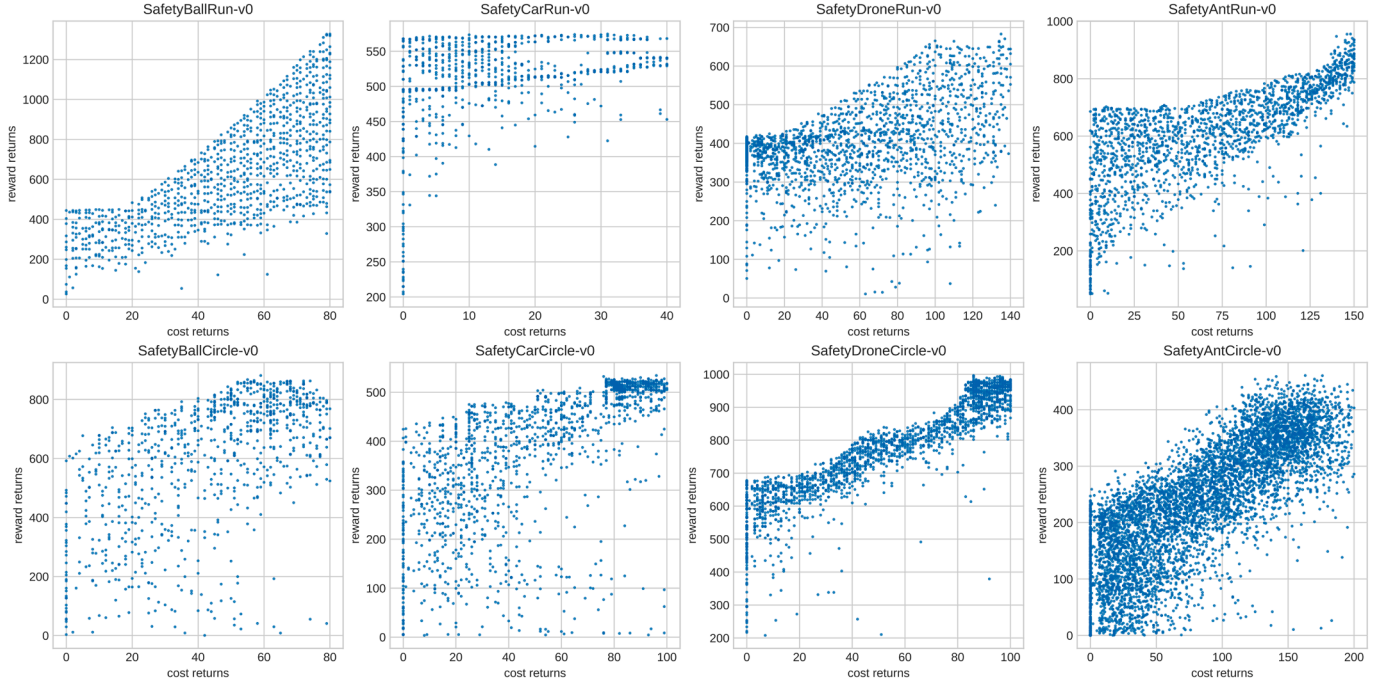


Fig. 2. Cost–Reward distribution on BulletSafetyGym.

1. Behavior Cloning (BC): BC learns a policy by directly copying the behavioral policy by supervised learning. Its goal is to imitate the demonstrated behaviors as accurately as it can.
2. CDT (Liu et al., 2023b): A Decision Transformer-based technique that integrates safety constraints into the decision-making process, leveraging the sequential structure of the offline data for improving risk-awareness.
3. BCQL: A Lagrangian-based adaptation of BCQ (Fujimoto et al., 2019) uses cost constraints to guide the learning process towards safe policies.
4. CPQ: This method treats OOD actions as unsafe and accordingly updates the Q-function using only safe state-action pairs, thereby promoting adherence to the defined safety boundary.
5. COptiDICE (Lee et al., 2022): An extension of OptiDICE (Lee et al., 2021) grounded in Distribution Correction Estimation (DICE) principles, designed specifically for safe offline RL by addressing distribution-shift and ensuring policy conformity to safety constraints.
6. FISOR (Zheng et al., 2024): A state-of-the-art OSRL algorithm that utilizes weighted diffusion regression to model safe trajectory distributions, offering a strong generative baseline for constraint satisfaction.

5.2. Comparative experiment analysis

Figs. 3 and 4 and Table 2 illustrate the performance comparison between OCPO-SA and all other comparative safe offline RL algorithms. In alignment with the DSRL benchmark's core requirements, we prioritize safety as the primary evaluation criterion, seeking to enhance reward performance while strictly adhering to safety thresholds. As shown in Figs. 3 and 4, OCPO-SA achieved 100% safety compliance across all tested environments, with the average cost reduced by 24% compared to the optimal baseline algorithm, which fully demonstrates the algorithm's safety. Additionally, OCPO-SA outperformed other baselines in 8 of 16 tested environments, demonstrating that the algorithm effectively balances policy safety with the utilization of offline reward information and thereby achieves a performance advantage.

Table 1
Hyper-parameters of OBPO-SA.

	Hyperparameter	Value
Basic	Optimizer	Adam
	Cost/Reward critic learning rate	1e-3
	Actor learning rate	1e-4
	Mini-batch size	512
	Discount factor	0.99
	Target update rate	5e-3
	Safe anchoring step	3e3
	Gradient step	1e6
Architecture	Cost/Reward critic hidden dimension	256
	Cost/Reward critic hidden layers	2
	Critic activation function	ReLU
	Actor hidden dimension	256
	Actor hidden layers	2
	Actor activation function	ReLU
Other	$[\alpha_1, \alpha_2]$	[2.5, 1]
	$[K_p, K_f, K_D]$	[0.1, 0.003, 0.001]

It should be noted that BCQL exhibits unstable training process and poor performance across multiple tasks due to the tripartite intertwined optimization problem (i.e., critic, actor, Lagrange multipliers). For BC, the performance and safety of learned policies are heavily influenced by the quality of the training dataset. We observe that BC unable to surpass the performance of the behavioral policy while also failing to ensure the safety of learned policies. As for BC_{safe}, although the algorithm can ensure safety, the overly conservative nature of the behavioral policy leads to unsatisfactory performance. Besides, COptiDICE satisfies the cost constraint by optimizing the steady-state distribution of the policy while maximizing the return. One can observe that in the CarGoal2 and CarPush1 tasks, the policy well satisfies the cost constraint, but fails to reach the best return. This may be because this algorithm is greatly affected by the offline data. Since CPQ adopts the conservative value estimation, the Q-function estimate is conservative and cannot represent the cost and the return accurately, resulting in poor performance in both performance and cost aspects. It can be seen from Table 2 that CDT performs comparably to our algorithm on several tasks. However, our algorithm

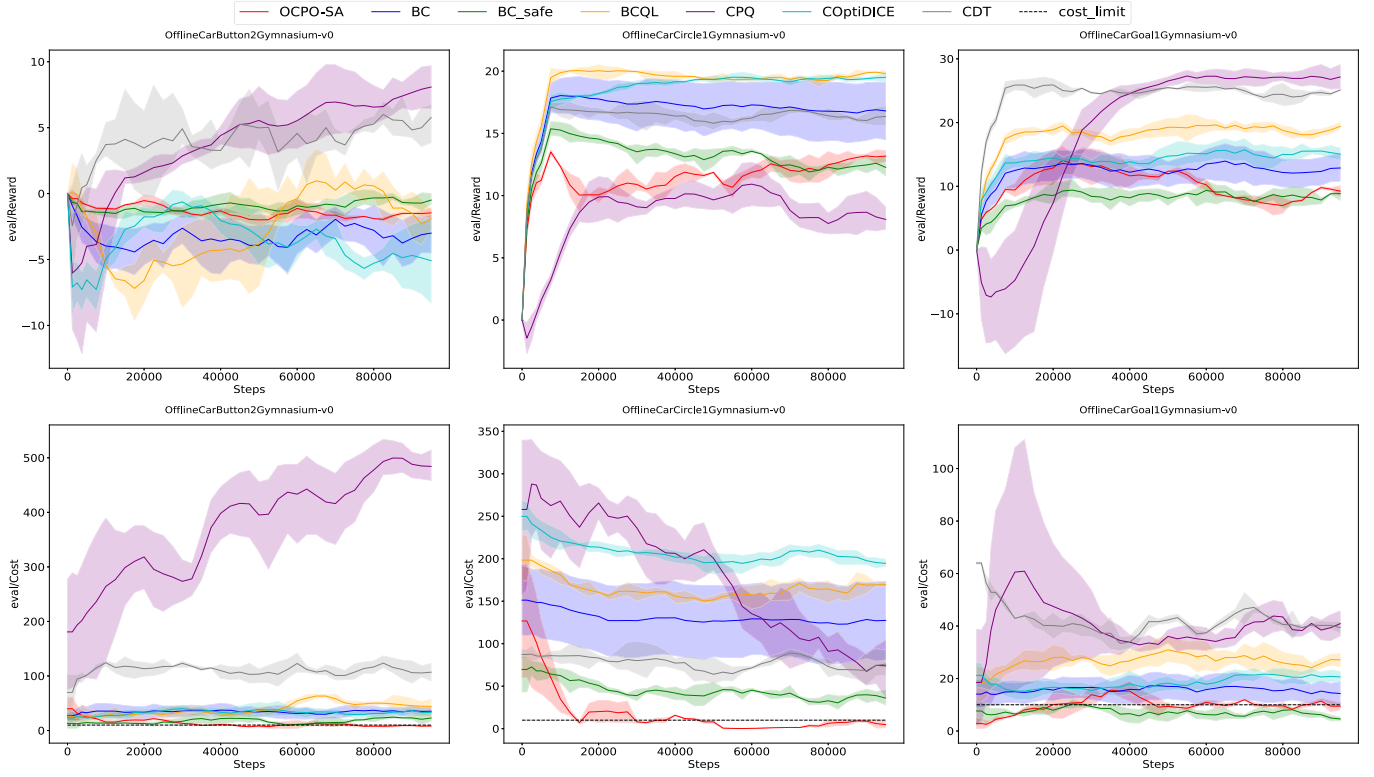


Fig. 3. Training results of various methods on the Safety-Gymnasium. The solid line represents the average return of evaluated algorithms over multiple random seeds, and the shaded area represents the standard deviation.

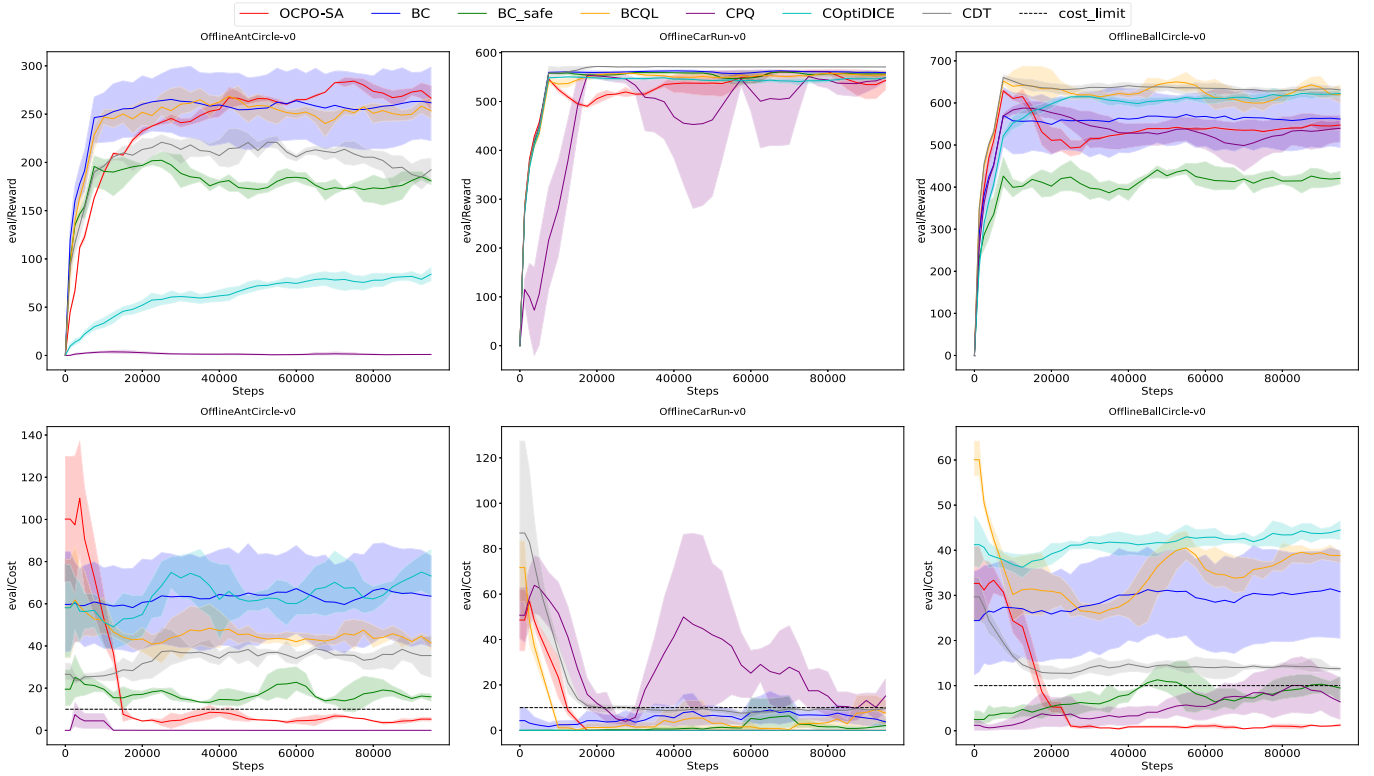


Fig. 4. Training results of various methods on the Bullet-Safety-Gym.

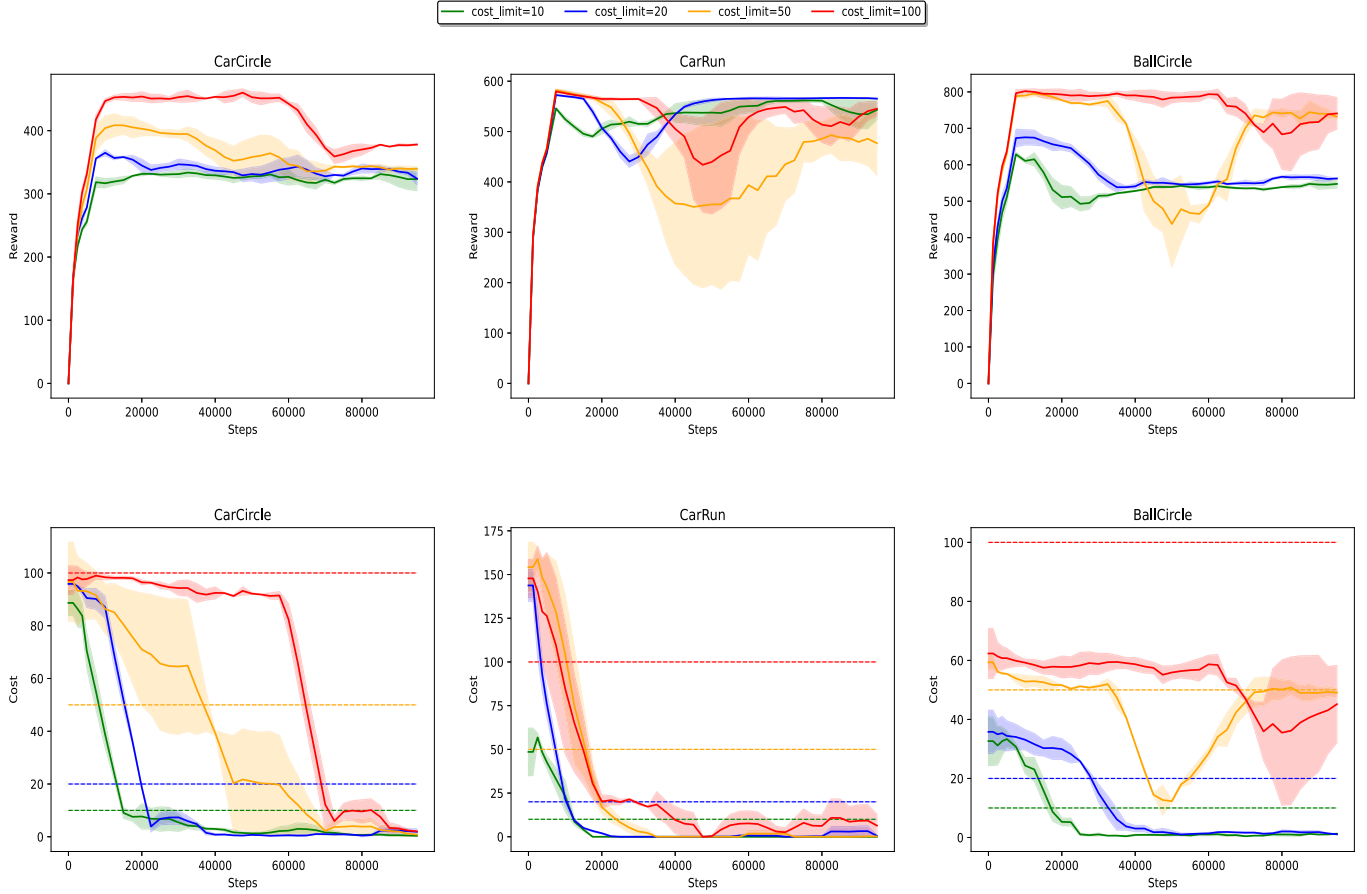


Fig. 5. Training results of OCPO-SA under different cost thresholds.

Table 2

Normalized DSRL (Liu et al., 2023a) benchmark results. $\uparrow$ stands for the higher the better. $\downarrow$ stands for the lower the better. Results are average of 10 eval episodes across 3 seeds. *Italic*: Unsafe agents. **Bold**: Safe agents with normalized cost smaller than 1. Underline: Safety agents with the maximum reward.

Task	BC		BC _{safe}		CDT		BCQ-Lag		CPQ		FISOR		COptiDICE		OCPO-SA (ours)	
	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$	reward $\uparrow$	cost $\downarrow$
CarButton1	0.03	1.38	0.00	0.06	0.21	1.6	0.04	1.63	0.42	9.66	-0.02	0.26	-0.08	1.68	0.05	0.58
CarButton2	-0.13	1.24	-0.01	0.63	0.13	1.58	0.06	2.13	0.37	12.51	0.01	0.58	-0.07	1.59	<u>0.04</u>	<u>0.38</u>
CarCircle1	0.72	4.39	0.19	1.52	0.6	1.73	0.73	5.25	0.02	2.29	0.10	0.13	0.7	5.72	<u>0.17</u>	<u>0.00</u>
CarCircle2	0.76	6.44	0.24	0.28	0.66	2.53	0.72	6.58	0.44	2.69	0.27	0.6	0.77	7.99	<u>0.34</u>	<u>0.00</u>
CarGoal1	0.39	0.33	0.24	0.28	0.66	1.21	0.47	0.78	0.79	1.42	<u>0.49</u>	<u>0.83</u>	0.35	0.54	0.44	0.89
CarGoal2	0.23	1.05	0.32	0.32	0.48	1.25	0.3	1.44	0.65	3.75	0.06	0.33	<u>0.25</u>	<u>0.91</u>	0.07	0.00
CarPush1	0.22	0.36	0.14	0.33	<u>0.31</u>	<u>0.4</u>	0.23	0.43	-0.03	0.95	0.28	0.28	0.23	0.5	0.18	0.84
CarPush2	<u>0.14</u>	<u>0.9</u>	0.05	0.45	<u>0.19</u>	1.3	0.15	1.38	0.24	4.25	0.14	0.89	0.09	1.07	0.06	0.60
BallRun	0.60	5.08	0.25	0.42	0.39	1.16	0.76	3.91	0.22	1.27	0.18	0.00	0.59	3.52	<u>0.30</u>	<u>0.00</u>
CarRun	0.97	0.33	0.97	0.00	<u>0.99</u>	<u>0.65</u>	0.94	0.15	0.95	1.79	0.73	0.14	0.87	0.00	0.97	0.00
DroneRun	0.24	2.13	0.07	0.98	0.63	0.79	0.72	5.54	0.33	3.52	0.30	0.55	0.67	4.15	<u>0.76</u>	<u>0.63</u>
AntRun	0.72	2.93	0.67	0.00	<u>0.72</u>	<u>0.91</u>	0.76	5.11	0.03	0.02	0.45	0.03	0.61	0.94	<u>0.21</u>	<u>0.09</u>
BallCircle	0.74	4.71	0.47	0.44	0.77	1.07	0.69	2.36	0.64	0.76	0.34	0.00	0.7	2.61	<u>0.68</u>	<u>0.74</u>
CarCircle	0.58	3.74	0.22	0.36	<u>0.75</u>	<u>0.95</u>	0.63	1.89	0.71	0.33	0.40	0.11	0.49	3.14	0.61	0.00
DroneCircle	0.72	3.03	0.45	0.00	<u>0.63</u>	<u>0.98</u>	0.8	3.07	-0.22	1.28	0.48	0.00	0.26	1.02	0.49	0.00
AntCircle	0.58	4.9	0.22	0.38	0.54	1.78	0.58	2.87	0.00	0.00	0.20	0.00	0.17	5.04	<u>0.62</u>	<u>0.16</u>

can ensure execution under absolute safety conditions across more tasks, even in the harder-to-control Gymnasium environment. FISOR employs a strict zero-violation algorithmic design that pursues extreme safety at the expense of reward maximization; thus, its return performance cannot be directly compared with OCPO-SA. Notably, generative methods such as CDT (which relies on the Transformer architecture) and FISOR

(which utilizes iterative diffusion steps) incur substantial computational and training costs. In contrast, our algorithm employs an MLP-based actor-critic architecture, achieving comparable or superior safe performance with fewer resources, thereby highlighting its superior efficiency in resource utilization. We will demonstrate this advantage in the ablation study.

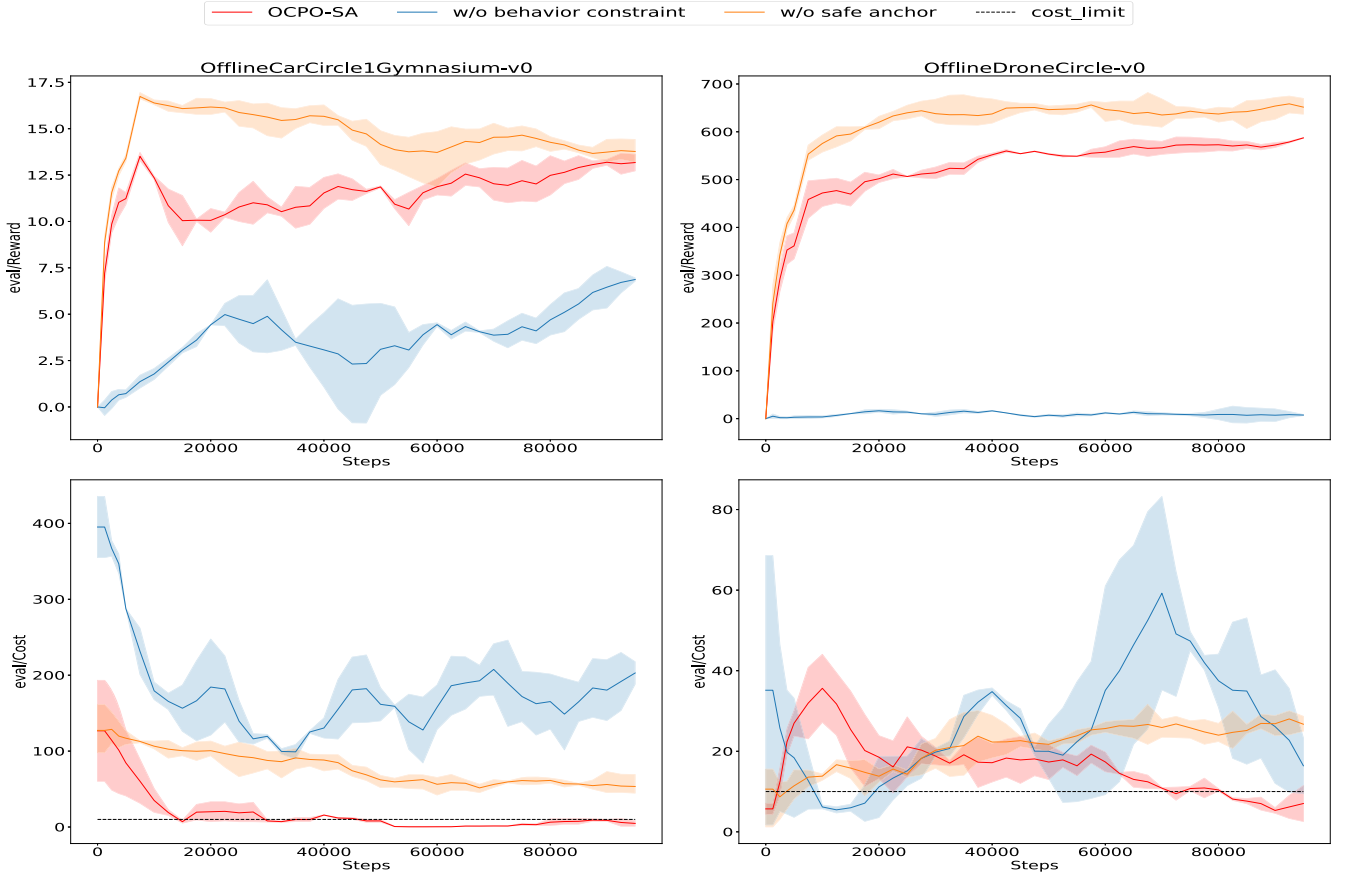


Fig. 6. Training results of OCPO-SA with and without behavioral constraint regularization and safe-anchor mechanism. “w/o behavior constraint” represents OCPO-SA without the behavioral constraint regularization term, and “w/o safe anchor” denotes OCPO-SA without the safe anchoring mechanism.

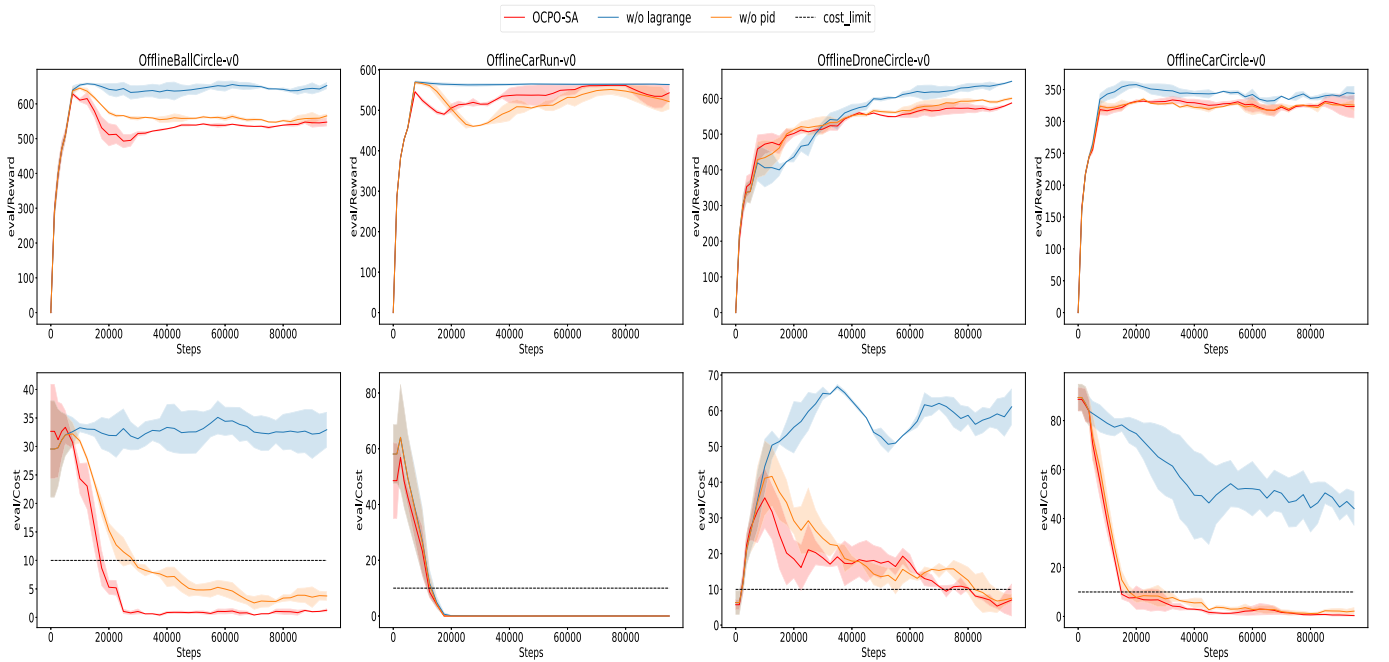


Fig. 7. Training results of OCPO-SA with and without lagrange regularization and pid lagrange regularization. “w/o lagrange” represents OCPO-SA without the lagrange regularization term, and “w/o pid” denotes OCPO-SA without lagrange regularization term.

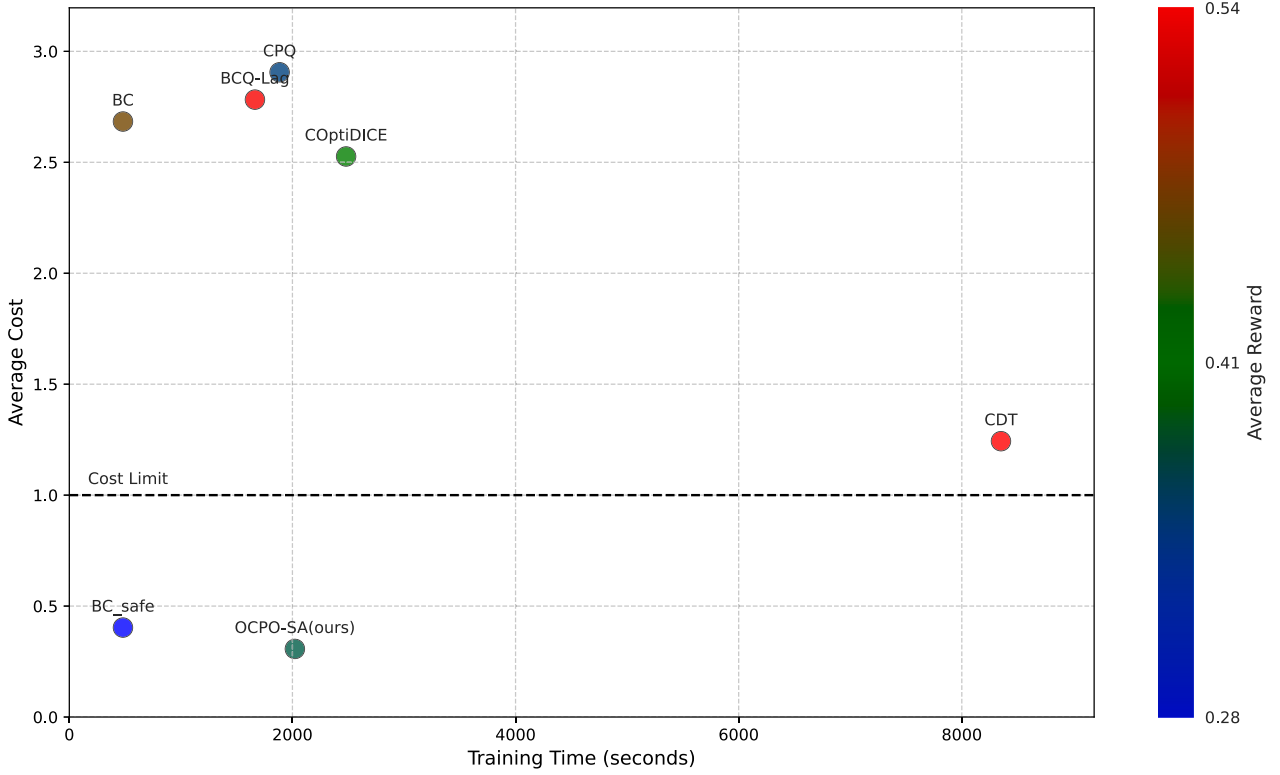


Fig. 8. Algorithm training complexity comparison: training time vs cost (color indicates reward).

5.3. Cost sensitivity analysis

To demonstrate the sensitivity of the proposed method to different cost thresholds, ablation experiments were conducted to evaluate its performance under varying cost thresholds. As shown in Fig. 5, the OCPO-SA algorithm exhibits differentiated performances across different cost thresholds (cost_limit = 10, 20, 50 and 100) in the CarCircle, CarRun, and BallCircle environments. In terms of reward, the curves under different cost thresholds in each environment show distinct trends: higher thresholds allow the policy to take more risks, leading to higher rewards. Regarding cost, the algorithm effectively drives the cost to converge under the set thresholds. Notably, under lower cost thresholds (e.g., 10 and 20), cost curves approach and stabilize near the corresponding thresholds more rapidly; while cost control under higher thresholds (e.g., 100) involves fluctuations, the algorithm still demonstrates adaptability and regulatory capabilities for cost constraints, collectively validating its performance characteristics in scenarios with different cost thresholds.

In the CarRun and CarCircle tasks, OCPO-SA maintains high rewards across various cost thresholds, with the learned policy's cost remaining at a relatively low level-attributed to the simplicity of these tasks.

5.4. Ablation study

5.4.1. Safe anchoring behavioral constraint analysis

According to the policy loss in Eq. (29), the policy network parameters are updated by optimizing a combined loss that comprises the Q-function, Lagrange constraint regularization term, and safe anchoring behavioral policy regularization item. Here, the safe anchoring behavioral policy regularization item serves to prevent excessive divergence between the policy's visiting distribution and safe behavioral policy. Next, we demonstrate the impact of the safe anchoring behavioral policy regularization on the policy optimization process. As shown in Fig. 6, experimental results indicate that in terms of performance (eval Reward) and cost (eval Cost), OCPO-SA and w/o safe anchoring with

behavioral policy constraint exhibit advantages over w/o behavior constraint (which lacks behavioral policy constraint). The former achieves better performance and cost control, while the latter not only shows poor performance but also maintains costs consistently above the cost limit. This stems from inaccurate estimations caused by distribution-shift. Due to the distortion of the value function and cost function, policy updates based on the value and cost function networks lead to a sharp decline in both performance and safety.

Further comparison reveals that when contrasting OCPO-SA with integrated safe anchoring behavioral constraint against w/o safe anchor (which adopts general behavioral policy constraint), OCPO-SA demonstrates more prominent cost control, stably maintaining costs at a lower level close to the cost limit, whereas w/o safe anchor incurs relatively higher costs. Meanwhile, OCPO-SA also exhibits more robust performance. According to Theorem 3, the performance upper bound of the trained policy is determined by the behavioral policy. The behavioral policy with general policy constraint uses trajectories containing unsafe behaviors, which leads to the insecurity of the trained policy.

5.4.2. Lagrange regularization analysis

The policy loss in Eq. (29) incorporates not only the behavioral policy regularization term but also the PID Lagrange constraint regularization term. The PID-Lagrange regularization term aims to ensure safety during the policy training process. In Fig. 7, we present the effects of general Lagrangian and PID-Lagrangian on the policy training process. In multiple benchmark tasks, the OCPO-SA without Lagrange regularization (w/o lagrange) significantly enhance performance (eval Reward), yet policy safety (eval Cost) declines sharply. This occurs because removing the Lagrange regularization term makes the policy optimization process neglect the safety constraints of the learned policy. Comparative analysis of PID-Lagrange (OCPO-SA) and standard Lagrange (w/o pid) demonstrated that the PID optimization achieved faster convergence rates in the OfflineBallCircle-v0 and OfflineDroneCircle-v0 environments. In the OfflineCarCircle-v0 environment, the impact of the two optimization methods on performance is negligible. This might be

because the fixed data distribution in simple scenarios makes the cost curve easy to converge. In the OfflineCarRun-v0 task, the presence or absence of the Lagrangian regularization term has no significant effect on policy training, which can be attributed to the task's simplicity, agents only need to navigate through a double-boundary channel rather than complex multi-obstacle scenarios, thus maintaining low costs even without Lagrangian regularization.

From the above experimental analysis, we conclude that dynamic PID adjustment of Lagrangian multipliers based on constraint functions can indeed enhance training stability and accelerate convergence, particularly in environments with non-trivial dynamics and data distributions where constraint violations are more likely to occur.

5.4.3. Training complexity analysis

To evaluate the training time efficiency of algorithms, we compared the average training duration across different algorithms. As shown in Fig. 8, within a similar training time frame, our OCPO-SA algorithm outperforms the actor-critic algorithms (CPQ, COptiDICE, and BCQ-lag) in terms of safety (quantified by average cost) and achieves comparable performance. The Transformer-based CDT algorithm, which incurs training complexity significantly higher than those of the OCPO-SA algorithm, offers high rewards but lacks sufficient safety. Although the training time of OCPO-SA is longer than that of the BC and BC_safe algorithms, it performs better in balancing the average reward and cost, highlighting its comprehensive advantage in balancing safety and efficiency.

6. Conclusion

We have proposed OCPO-SA, a novel safe offline RL algorithm. To ensure safety and address distribution-shift, we incorporate cumulative cost constraints and behavioral regularization into the policy optimization objective and derive an analytical solution to the resulting constrained policy improvement problem. We further analyze monotonic performance improvement and provide safety guarantees in terms of bounded expected costs under the offline dataset distribution. In practical implementation, we employ a VAE-distilled safe anchoring mechanism to restrict policy updates within a distilled safe action space and adopt an adaptive Lagrangian scheme with PID-style updates for stable constraint satisfaction. Experimental results on Safety-Gymnasium and Bullet-Safety-Gym demonstrate that OCPO-SA achieves strong performance while maintaining safety compliance across all tested environments. In this work, “maintaining safety” should be interpreted as achieving constraint satisfaction under the benchmark protocol, rather than guaranteeing zero violations in all possible deployments.

Limitations include (i) scalability to higher-dimensional continuous-control tasks where modeling the safe action space may require larger generative models, (ii) dependence on dataset coverage and safety distribution, and (iii) inherent reward–cost trade-offs controlled by constraint and regularization hyperparameters. Future work will explore stronger notions of safety (e.g., stricter constraints) and more scalable safe generative models to address higher-dimensional continuous-control.

CRedit authorship contribution statement

DiYuan Hou: Writing – review & editing, Writing – original draft, Software, Methodology; **Longyang Huang:** Writing – review & editing, Methodology; **Pu Feng:** Writing – review & editing, Methodology; **Wenjun Wu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Key R&D Program of China (2025YFF1505704) and in part by the National Key Laboratory of Complex Critical Software Environment under grant SKLCCSE-2025ZX-01.

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